

CS 6371 - Assignment 8

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$$1. (\forall \alpha. (\alpha \times \alpha)) + (\forall \alpha. (\alpha \rightarrow \text{unit}))$$

$$\mathbf{in}_2^{(\forall \alpha. (\alpha \times \alpha)) + (\forall \alpha. (\alpha \rightarrow \text{unit}))} (\Lambda \alpha. \lambda x: \alpha. ())$$

$$2. \forall \alpha. ((\alpha \times \alpha) + (\alpha \rightarrow \text{unit}))$$

$$\Lambda \alpha. (\mathbf{in}_2^{(\alpha \times \alpha) + (\alpha \rightarrow \text{unit})} (\lambda x: \alpha. ()))$$

$$3. \forall \alpha. \forall \beta. ((\alpha + \beta) \rightarrow ((\alpha \times \text{int}) + (\text{bool} \times \beta)))$$

$$\Lambda \alpha. \Lambda \beta. \left(\lambda x: \alpha + \beta. \left(\mathbf{case } x \mathbf{ of } \mathbf{in}_1^{\alpha + \beta}(y) \rightarrow \mathbf{in}_1^{(\alpha \times \text{int}) + (\text{bool} \times \beta)}((y, 3)) \right. \right. \\ \left. \left. \mathbf{in}_2^{\alpha + \beta}(z) \rightarrow \mathbf{in}_2^{(\alpha \times \text{int}) + (\text{bool} \times \beta)}((\text{true}, z)) \right) \right)$$

$$4. \forall \beta. (\text{void} \rightarrow \beta)$$

$$\Lambda \beta. \lambda x: \text{void}. (x[\beta])$$

$$5. \tau = \forall \alpha. \forall \beta. ((\alpha \rightarrow \beta) \rightarrow (\beta \rightarrow \alpha))$$

$$\begin{aligned} & \mathcal{I}(\forall \alpha. \forall \beta. ((\alpha \rightarrow \beta) \rightarrow (\beta \rightarrow \alpha))) \\ &= \forall \alpha: \text{bool}, \mathcal{I}(\forall \beta. ((\alpha \rightarrow \beta) \rightarrow (\beta \rightarrow \alpha))) \\ &= \forall \alpha: \text{bool}, \forall \beta: \text{bool}, \mathcal{I}((\alpha \rightarrow \beta) \rightarrow (\beta \rightarrow \alpha)) \\ &= \forall \alpha: \text{bool}, \forall \beta: \text{bool}, \mathcal{I}(\alpha \rightarrow \beta) \Rightarrow \mathcal{I}(\beta \rightarrow \alpha) \\ &= \forall \alpha: \text{bool}, \forall \beta: \text{bool}, (\alpha \Rightarrow \beta) \Rightarrow (\beta \Rightarrow \alpha) \end{aligned}$$

This property does not hold for $\alpha = F$ and $\beta = T$, therefore $\mathcal{I}(\tau) = F$.

$$\begin{aligned}
& \forall \alpha:bool, \forall \beta:bool, (\alpha \Rightarrow \beta) \Rightarrow (\beta \Rightarrow \alpha) \\
& \quad (F \Rightarrow T) \Rightarrow (T \Rightarrow F) \\
& = T \Rightarrow F \\
& = F
\end{aligned}$$

Thus, we conclude that τ is uninhabited.